

LAB 5

Task # 01: Raising a number n to a power p is the same as multiplying n by itself p times. Write a function called `power()` that takes a value for n and an int value for p , and returns the result as a double value.

CODE:

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        String[] names = new String[10];
        int[] marks = new int[10];

        // Input names and marks of 10 students
        for (int i = 0; i < 10; i++) {
            System.out.print("Enter name of student " + (i + 1) + ": ");
            names[i] = scanner.nextLine();
            System.out.print("Enter marks of " + names[i] + ": ");
            marks[i] = scanner.nextInt();
            scanner.nextLine(); // consume the newline character left by nextInt()
        }

        // Initialize variables to track highest and lowest marks
        int highestMarks = marks[0];
        int lowestMarks = marks[0];
        String highestStudent = names[0];
        String lowestStudent = names[0];

        // Find the highest and lowest marks and corresponding students
        for (int i = 1; i < 10; i++) {
            if (marks[i] > highestMarks) {
                highestMarks = marks[i];
                highestStudent = names[i];
            }
            if (marks[i] < lowestMarks) {
                lowestMarks = marks[i];
                lowestStudent = names[i];
            }
        }

        // Output the results
        System.out.println("\nStudent with highest marks: " + highestStudent + " with marks: " + highestMarks);
        System.out.println("Student with lowest marks: " + lowestStudent + " with marks: " + lowestMarks);

        scanner.close();
    }
}
```

```
Enter name of student 1: Muneeb
Enter marks of Muneeb: 85
Enter name of student 2: Muneeb
Enter marks of Muneeb: 85
Enter name of student 3: Muneeb
Enter marks of Muneeb: 85
Enter name of student 4: Muneeb
Enter marks of Muneeb: 85
Enter name of student 5: Muneeb
Enter marks of Muneeb: 85
Enter name of student 6: Muneeb
Enter marks of Muneeb: 85
Enter name of student 7: Muneeb
Enter marks of Muneeb: 85
Enter name of student 8: Muneeb
Enter marks of Muneeb: 85
Enter name of student 9: Muneeb
Enter marks of Muneeb: 0
Enter name of student 10: Muneeb
Enter marks of Muneeb: 100

Student with highest marks: Muneeb with marks: 100
Student with lowest marks: Muneeb with marks: 0
```

LAB 5

Task # 02: Raising a number n to a power p is the same as multiplying n by itself p times. Write a function called `power()` that takes a value for n and an int value for p , and returns the result as a double value.

CODE:

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter your name: ");
        String name = scanner.nextLine();

        name = name.toLowerCase();

        int consonantCount = 0;

        for (int i = 0; i < name.length(); i++) {
            char ch = name.charAt(i);

            if (Character.isLetter(ch) && !isVowel(ch)) {
                consonantCount++;
            }
        }
        System.out.println("Total number of consonants in your name: " + consonantCount);
        scanner.close();
    }
    public static boolean isVowel(char ch) {
        return (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u');
    }
}
```

OUTPUT:

```
Enter your name: Muneeb
Total number of consonants in your name: 3
```

LAB 5

Task # 03: Raising a number n to a power p is the same as multiplying n by itself p times. Write a function called power() that takes a value for n and an int value for p , and returns the result as a double value.

CODE:

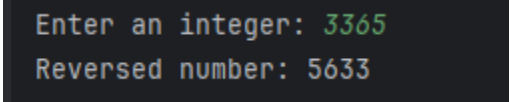
```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter an integer: ");
        int num = scanner.nextInt();

        int reversed = 0;

        while (num != 0) {
            int digit = num % 10;
            reversed = reversed * 10 + digit;
            num = num / 10;
        }
        System.out.println("Reversed number: " + reversed);

        scanner.close();
    }
}
```

OUTPUT:

A screenshot of a terminal window with a dark background. It shows the output of the program: "Enter an integer: 3365" followed by "Reversed number: 5633".

```
Enter an integer: 3365
Reversed number: 5633
```

Task # 04: Write a java program which reads a string from the keyboard and prints its words in reverse order. Following is the sample run of the program: Enter text: Welcome to Iqra University Reverse: University Iqra to Welcome.

CODE:

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter an integer: ");
        int num = scanner.nextInt();

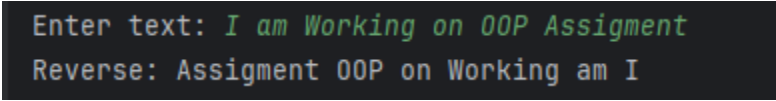
        int reversed = 0;
```

LAB 5

```
while (num != 0) {
    int digit = num % 10;
    reversed = reversed * 10 + digit;
    num = num / 10;
}
System.out.println("Reversed number: " + reversed);

scanner.close();
}
}
```

OUTPUT:



Enter text: *I am Working on OOP Assigment*
Reverse: Assignment OOP on Working am I

Task # 05: Creating and Populating an Array of your choice.

CODE:

```
public class Main {
    public static void main(String[] args) {
        int[] numbers = {4, 7, 2, 9, 5};
        System.out.print("Array elements: ");
        for (int num : numbers) {
            System.out.print(num + " ");
        }
        System.out.println();

        int sum = 0;
        for (int num : numbers) {
            sum += num;
        }
        System.out.println("Sum: " + sum);

        int largest = numbers[0], smallest = numbers[0];
        for (int num : numbers) {
            if (num > largest) largest = num;
            if (num < smallest) smallest = num;
        }
        System.out.println("Largest: " + largest);
        System.out.println("Smallest: " + smallest);

        System.out.print("Squares: ");
        for (int num : numbers) {
            System.out.print((num * num) + " ");
        }
    }
}
```

LAB 5

```
System.out.println();

String str = "Hello";
System.out.print("Characters: ");
for (int i = 0; i < str.length(); i++) {
    System.out.print(str.charAt(i) + " ");
}
System.out.println();
}
}
```

CODE:

```
Array elements: 4 7 2 9 5
Sum: 27
Largest: 9
Smallest: 2
Squares: 16 49 4 81 25
Characters: H e l l o
```